



Computational and Differential Geometry

Euler spiral or Clothoid curve

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Consider the parametrization of the Euler spiral given by:

$$x(s) = \int_0^s \cos(l^2) dl, \quad y(s) = \int_0^s \sin(l^2) dl$$

such that the curve is parametrized by $\alpha(s) = (x(s), y(s), 0)$.

1. Is the curve parameterized by arc length?
2. Plot the functions for the interval $-10 < s < 10$.
3. Plot the curve $\alpha(s)$ for $-10 < s < 10$.
4. Evaluate and plot the curvature $\kappa(s)$ for the interval $-10 < s < 10$.
5. Evaluate the binormal vector $\hat{B}(s)$ and interpret the result.

Hints: The integral $\int_0^s x dx = \frac{s^2}{2}$ is evaluated as

```
x, s = sp.symbols("x, s", real = True)
sp.integrate(x, (x, 0, s))
```